Randy Zhu

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Education

University of British Columbia April 2028
Bachelor of Science, Computer Science, Option in Software Engineering *GPA: 3.86/4*
Relevant Coursework: Compilers (A+), Programming Languages (A+), Computer Architecture and OS (A-)

Technical Skills

Languages: C, C++, Rust, Swift, C#, Java, Python, SQL, TypeScript, HTML+CSS, Racket
Compilers: LLVM IR, SSA form, dataflow analysis
Systems & Hardware: Darwin Kernel, device trees, ARM
Tools: GDB, Valgrind, Sanitizers, perf, dtrace, CMake, cross-compilation, QEMU

Experience

Firmware Engineer Intern May 2026 – Present
Netgear *Vancouver, BC*

- Point 1
  extra stuff for p1
- Point 2
- Point 3
  extra stuff for p3
- Point 4
- Point 5
  extra stuff for p5

Research Intern May 2025 – September 2025
Software Practices Lab (UBC) *Vancouver, BC*

- Secured $6000 grant to conduct full-time research, directly supervised by Dr. Caroline Lemieux, a Canada Research Chair
- Created tool which works on 64% of popular open-source repositories that generates unit tests from integration tests
- Implemented various LLM guided control and data flow analyses that integrated with Gemma 3n, an on device model

Software Developer Intern September 2024 – April 2025
Teck Resources *Vancouver, BC*

- Saved 100 hours for mine site engineering teams by creating a data ingest tool using the Dataverse API in C#
- Introduced team to Playwright, catching 87% of bugs before reaching manual testing using Playwright and NUnit
- Created calendar component used by teams across the org using React.js, TypeScript and Fluent UI React toolkit

Teaching Assistant July 2024 – Present
University of British Columbia *Vancouver, BC*

- Covered professor's lectures on MIPS assembly, stack frames, and POSIX threading to 200+ students
- Achieved 98% favourable rating (n=102); helped students debug concurrent C code using the pthread library

Projects

macOS iSCSI Driver | *C++, DriverKit, IOKit, Swift* November 2025 – Present

- iSCSI is a storage protocol similar to Windows File Sharing (SMB) or NFS, but at a lower level. This project aims to create an free and open-source alternative that uses the new DriverKit API instead of kernel modules.
- Working around limitations of DriverKit by implementing the networking client in user space with Swift
- Improved performance by designing and using a ring buffer to forward driver SCSI data to user space

Racketish Compiler | *x86_64 Assembly, LLVM IR, gdb, Racket* January 2026 – May 2026

- Reduced compile times 180x (3h $\rightarrow$ 1m) by eliminating a pathological typed/untyped bottleneck in Typed Racket
- Replaced x86-only codegen with an LLVM IR backend, enabling compilation to any LLVM-supported platform
- Implemented closure conversion, tagged value representation, and tail call optimization via jump conversion

Sasquatch | *Swift, ARKit, LiDAR, Detectron2, Open3D, Python, FastAPI, PostgreSQL, GCP*

- Built an ML pipeline combining Detectron2 instance segmentation with LiDAR point cloud processing to detect climbing wall holds and generate routes of varying difficulty
- Implemented PCA-based wall orientation detection to automatically position a virtual pinhole camera, projecting 3D point clouds to 2D for inference and back-projecting detections to world coordinates
- Integrated Google Gemini for hold classification, Open3D for 3D processing, and FastAPI for a real-time API consumed by an iOS capture app implemented in Swift with SwiftUI